Trigonométrie — formulaire express

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	—	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0

Identités & périodicité

$$\cos^2 x + \sin^2 x = 1, \qquad 1 + \tan^2 x = \frac{1}{\cos^2 x}$$

$\tan x = \frac{\sin x}{\cos x}$; $\cos, \sin$ sont 2π-pér., $\tan$ π-pér. ; $\cos$ paire, $\sin, \tan$ impaires.

Angles associés

$$\begin{array}{ll}
\cos(-x) = \cos x & \sin(-x) = -\sin x \\
\cos(\pi - x) = -\cos x & \sin(\pi - x) = \sin x \\
\cos(\pi + x) = -\cos x & \sin(\pi + x) = -\sin x \\
\cos(\frac{\pi}{2} - x) = \sin x & \sin(\frac{\pi}{2} - x) = \cos x \\
\cos(\frac{\pi}{2} + x) = -\sin x & \sin(\frac{\pi}{2} + x) = \cos x
\end{array}$$

Équations de référence

$$\cos x = \cos \alpha \iff x = \pm\alpha + 2k\pi$$
$$\sin x = \sin \alpha \iff x = \alpha + 2k\pi \text{ ou } \pi - \alpha + 2k\pi$$
$$\tan x = \tan \alpha \iff x = \alpha + k\pi \quad (k \in \mathbb{Z})$$

Addition

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$
$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$
$$\tan(a \pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}$$

Duplication & carré

$$\cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$$
$$\sin 2x = 2 \sin x \cos x, \qquad \tan 2x = \frac{2\tan x}{1 - \tan^2 x}$$
$$\cos^2 x = \frac{1 + \cos 2x}{2}, \qquad \sin^2 x = \frac{1 - \cos 2x}{2}$$

Linéarisation

$$\cos^3 x = \frac{3}{4}\cos x + \frac{1}{4}\cos 3x$$
$$\sin^3 x = \frac{3}{4}\sin x - \frac{1}{4}\sin 3x$$

Angle moitié $t = \tan \frac{x}{2}$

$$\cos x = \frac{1-t^2}{1+t^2}, \ \sin x = \frac{2t}{1+t^2}, \ \tan x = \frac{2t}{1-t^2}$$

Somme $\to$ produit

$$\cos p + \cos q = 2\cos\frac{p+q}{2}\cos\frac{p-q}{2}$$
$$\cos p - \cos q = -2\sin\frac{p+q}{2}\sin\frac{p-q}{2}$$
$$\sin p + \sin q = 2\sin\frac{p+q}{2}\cos\frac{p-q}{2}$$
$$\sin p - \sin q = 2\cos\frac{p+q}{2}\sin\frac{p-q}{2}$$

Produit $\to$ somme

$$\cos a \cos b = \frac{1}{2}[\cos(a+b) + \cos(a-b)]$$
$$\sin a \sin b = \frac{1}{2}[\cos(a-b) - \cos(a+b)]$$
$$\sin a \cos b = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$$

$a\cos x + b \sin x$

$$a\cos x + b\sin x = \sqrt{a^2+b^2}\,\cos(x - \varphi)$$

$\cos\varphi = \frac{a}{\sqrt{a^2+b^2}}$, $\sin\varphi = \frac{b}{\sqrt{a^2+b^2}}$; donc $|a\cos x + b\sin x| \le \sqrt{a^2+b^2}$.

Euler & Moivre (CPGE)

$$e^{\mathrm{i}x} = \cos x + \mathrm{i}\sin x, \ (\cos x + \mathrm{i}\sin x)^n = \cos nx + \mathrm{i}\sin nx$$
$$\cos x = \frac{e^{\mathrm{i}x} + e^{-\mathrm{i}x}}{2}, \qquad \sin x = \frac{e^{\mathrm{i}x} - e^{-\mathrm{i}x}}{2\mathrm{i}}$$